
VERIFIED PEPTIDES NETWORK

Peptide Storage, Reconstitution & Solvent Guide

Storage Temperatures · Shelf Life · Reconstitution Step-by-Step · Solvent Selection ·
Peptide-by-Peptide Reference

BAC WATER · ACETIC ACID · GEL & DIFFICULT PEPTIDES · STABILITY DATA · SAFETY RULES

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1. Safety & Research Disclaimer

■ All content is for research and educational purposes only.

Peptides referenced in this guide are research chemicals intended exclusively for in-vitro laboratory research and scientific investigation. They are NOT intended, approved, or marketed for human consumption. Always work in a clean environment, use sterile technique, and verify your compounds with independent COA testing before any research use.

2. The Golden Rules of Peptide Storage

Keep it cold	Temperature is the single biggest factor in peptide stability. Lyophilized peptides stored at room temperature can degrade in days to weeks. Refrigerator extends stability to months. Freezer (-20 °C) extends to 1–2+ years.
Keep it dry	Moisture is the enemy of lyophilized peptide powder. Never open a vial in a humid environment. Use desiccant storage bags or airtight containers.
Keep it dark	UV light degrades many peptides — particularly those with aromatic amino acids (Trp, Tyr, Phe). Store in opaque containers or amber vials, away from direct light at all times.
Never shake	Peptides are delicate molecular structures. Shaking can cause aggregation and denaturation. When mixing reconstituted peptide, gently swirl or roll the vial between your palms. Never vortex.
Reconstitute only what you need	Once reconstituted, peptide stability drops significantly. Do not reconstitute large vials months in advance. Reconstitute one vial at a time and use within the stability window.
Label everything	Write the peptide name, date of reconstitution, concentration, and solvent on every vial. Use a Niimbot label printer or waterproof marker. Memory is not a storage system.
Use bacteriostatic water for multi-dose vials	BAC water contains 0.9% benzyl alcohol — a bacteriostatic agent that prevents microbial growth across multiple draws. Plain sterile water should only be used for single-dose vials consumed immediately.

3. Lyophilized (Dry Powder) — Shelf Life & Conditions

Lyophilized (freeze-dried) peptide powder is the most stable form. The powder has been vacuum-dried to remove moisture, making it far less vulnerable to hydrolysis than reconstituted peptide. Proper storage dramatically extends usable life.

CONDITION	STABILITY	NOTES
Room temperature (18–25°C)	Days to 2–4 weeks	Not recommended. Short-term transit only.
Refrigerator (2–8°C)	3–6 months (most peptides)	Acceptable for regular-use vials. Keep sealed with desiccant.
Freezer (-20°C)	1–2 years (most peptides)	Best for long-term storage. Allow to warm to room temp before opening to avoid condensation.
Deep freeze (-80°C)	2–5+ years	Research-grade long-term archival. Overkill for most home research protocols.

Important pre-storage rules:

- Allow the vial to reach room temperature BEFORE opening. If you open a cold vial, warm humid air rushes in and condenses moisture into the powder — this starts degradation immediately.
- Seal opened but unreconstituted vials with parafilm or PTFE tape, then return to freezer immediately.
- Do not puncture the rubber stopper until ready to reconstitute — the sealed vial is airtight and the stopper maintains sterility.
- Store upright to keep powder away from the stopper.
- Write the date received on the vial at arrival — track from that date, not the date you open it.

4. Reconstituted Peptides — How Long They Last

Once a peptide is dissolved in solvent, the stability clock resets and runs much faster. Water molecules attack the peptide bonds (hydrolysis). Bacterial contamination risk increases with every needle puncture. The type of solvent used matters enormously — BAC water extends reconstituted shelf life; plain sterile water does not.

CONDITION	STABILITY WITH BAC WATER	STABILITY WITH STERILE WATER
Refrigerator (2–8°C)	4–6 weeks for most peptides	5–7 days maximum
Freezer (-20°C)	3–6 months (some peptides 1 year)	1–3 months (microbial risk removed but hydrolysis continues)
Room temperature	24–48 hours only	12–24 hours only

■ Peptides That Degrade Faster After Reconstitution:

GHK-Cu: Copper chelation can accelerate oxidation. Use within 2–3 weeks refrigerated with BAC water.

PT-141 (Bremelanotide): Relatively sensitive to hydrolysis. Use within 4 weeks refrigerated.

Melanotan I & II: Sensitive to light and oxidation post-reconstitution. Keep in amber vial. Use within 4 weeks.

Kisspeptin: Short stability window. Use within 2–3 weeks refrigerated.

Oxytocin: Degrades faster than most peptides. Use within 2–3 weeks. Avoid repeated freeze-thaw.

LL-37: Antimicrobial peptides can self-associate. Use within 3–4 weeks refrigerated.

SS-31 (Elamipretide): Oxidation-sensitive. Limit light exposure. Use within 2–4 weeks refrigerated.

More Stable After Reconstitution:

BPC-157: 4–6 weeks refrigerated with BAC water. Some researchers report 8+ weeks with proper storage.

TB-500: 4–6 weeks refrigerated. Relatively robust stability profile.

CJC-1295 DAC: 4–6 weeks refrigerated. DAC modification increases stability.

Semaglutide: 4–6 weeks refrigerated (mirrors pharmaceutical product labeling).

Tirzepatide: 4–6 weeks refrigerated. Match pharmaceutical analogue guidelines.

Tesamorelin: 4–6 weeks refrigerated with BAC water.

Sermorelin: 4–6 weeks refrigerated. Well-studied stability profile.

Epitalon: 4–6 weeks refrigerated. Khavinson peptides generally stable.

NAD+: Use immediately or within 24 hours of reconstitution for IV; NMN/NR oral forms have long shelf life in powder form.

5. Solvent Selection — BAC Water vs Acetic Acid vs Other

Choosing the wrong solvent is one of the most common reconstitution errors. Most peptides dissolve well in bacteriostatic water (BAC water). A subset of peptides — particularly those with many hydrophobic residues, or those containing cysteine or methionine — require dilute acetic acid to dissolve. Some gel-like or insoluble peptides need special handling.

SOLVENT	WHAT IT IS	WHEN TO USE	AMOUNT TO ADD
Bacteriostatic Water (BAC Water)	Sterile water for injection containing 0.9% benzyl alcohol as a bacteriostatic preservative	Default solvent for most peptides. Use for all multi-dose vials. Do not use plain sterile water for multi-dose.	1–2 mL per 5 mg vial (standard). Adjust to desired concentration.
Sterile Water for Injection	Purified water, no preservatives. Single-use only.	Single-dose vials consumed immediately. Some peptides sensitive to benzyl alcohol (rare).	Same as BAC water. Discard remaining within 24 hours.
0.6% Acetic Acid (Dilute Acetic Acid)	White vinegar-grade diluted in sterile water. Lowers pH to help dissolve hydrophobic or difficult peptides.	Peptides that do not dissolve in BAC water. Particularly: IGF-1, IGF-1 LR3, MGF, GHRP variants, Hexarelin, and others listed below.	0.5–1 mL of 0.6% acetic acid to dissolve, then dilute with BAC water to final volume.
DMSO (Dimethyl Sulfoxide)	Polar aprotic solvent. High penetration ability. Pharmaceutical-grade only.	Very rare. Only for research peptides that are completely insoluble in all aqueous solvents. Not for IV/SC use.	Use minimum volume possible. Always dilute heavily with sterile saline before any use.
Propylene Glycol (PG)	Viscous polar solvent. Common in topical and subcutaneous formulations.	Topical peptide formulations. Occasionally used for PT-141 nasal sprays.	10–20% PG in sterile water for topical. Follow vendor formulation instructions.

How to Make 0.6% Acetic Acid Solution:

You cannot use household white vinegar directly (5% acetic acid — too concentrated and not sterile). The correct method:

1. Purchase pharmaceutical-grade glacial acetic acid (99%+) or use food-grade acetic acid.
2. Mix 6 mL of glacial acetic acid into 994 mL of sterile water for injection = 0.6% solution.
3. Alternatively: mix 1.2 mL glacial acetic acid into 198.8 mL sterile water = 200 mL of 0.6%.
4. Filter through a 0.22 micron syringe filter before use.
5. Store refrigerated in a sealed container. Label: '0.6% Acetic Acid Solution — Date Made'.
6. Use within 1–3 months.

6. Peptide-by-Peptide Solvent & Reconstitution Reference

This reference table lists the recommended solvent, standard reconstitution volume, storage conditions, and special notes for each major research peptide. When in doubt, attempt BAC water first — if the peptide does not dissolve after gentle rolling for 15–20 minutes, switch to acetic acid protocol.

PEPTIDE	SOLVENT	RECONSTITUTION	REFRIGERATOR SHELF LIFE	NOTES
BPC-157	BAC Water	1–2 mL per 5 mg	4–6 weeks	Most versatile peptide. Oral use: dissolve in sterile water, drink immediately.
TB-500	BAC Water	1–2 mL per 5 mg	4–6 weeks	Dissolves easily. Often comes as 2 mg or 5 mg vials.
KPV	BAC Water	1–2 mL per 5 mg	4–6 weeks	Oral: dissolve in sterile water and consume within 24 hours.
GHK-Cu	BAC Water	1–2 mL per 5 mg	2–3 weeks	Blue-green color when dissolved is normal (copper complex). Keep dark.
AHK-Cu	BAC Water or Propylene Glycol	1 mL per 1 mg (topical: PG 20%)	3–4 weeks (topical: 2 months)	Topical formulation common. For SC: BAC water standard.
Semax	BAC Water	1–2 mL per 1 mg	3–4 weeks	For nasal spray: 1 mg/mL in BAC water. Use nasal atomizer.
Na Semax	BAC Water	1–2 mL per 1 mg	3–4 weeks	Same as Semax. Slightly more potent per volume.
Selank	BAC Water	1–2 mL per 1 mg	3–4 weeks	Nasal use standard. 1 mg/mL solution common.
Na Selank	BAC Water	1–2 mL per 1 mg	3–4 weeks	Longer-acting variant. Same reconstitution approach as Selank.
CJC-1295 DAC	BAC Water	1–2 mL per 2 mg	4–6 weeks	Dissolves readily. Weekly dosing makes good use of 4–6 week window.
CJC-1295 No DAC	BAC Water	1–2 mL per 2 mg	4–6 weeks	Often pre-mixed with Ipamorelin. Stable in BAC water.
Ipamorelin	BAC Water	1–2 mL per 2 mg	4–6 weeks	Clean and easy to reconstitute. No special solvent required.
GHRP-2	BAC Water or 0.6% Acetic Acid	1–2 mL per 5 mg	3–4 weeks	Usually dissolves in BAC water. If not, use dilute acetic acid.
GHRP-6	BAC Water or 0.6% Acetic Acid	1–2 mL per 5 mg	3–4 weeks	Same as GHRP-2. Start with BAC water.
Hexarelin	0.6% Acetic Acid then BAC Water	0.5 mL AA, then 1–1.5 mL BAC	3–4 weeks	More hydrophobic than other GHRPs. Acetic acid first is recommended.
Sermorelin	BAC Water	1–2 mL per 3 mg	4–6 weeks	Stable in BAC water. Commonly supplied as 3 mg or 6 mg vials.

Tesamorelin	BAC Water	1–2 mL per 1–2 mg	4–6 weeks	FDA-approved compound. Dissolves readily. Store reconstituted refrigerated.
Semaglutide	BAC Water	1–2 mL per 3–10 mg	4–6 weeks	Long half-life (7 days). Entire vial typically reconstituted at once for weekly dosing.
Tirzepatide	BAC Water	1–2 mL per 3–15 mg	4–6 weeks	Same approach as semaglutide. Concentrate to desired dose per draw.
Retatrutide	BAC Water	1–2 mL per 5–10 mg	4–6 weeks	Dissolves readily. Adjust concentration for precise weekly draw volumes.
Cagrilintide	BAC Water	1–2 mL per 5 mg	4–6 weeks	Amylin analog. Same BAC water protocol as other GLP class compounds.
AOD-9604	BAC Water	1–2 mL per 5 mg	4–6 weeks	HGH fragment. Dissolves easily in BAC water.
HGH Fragment 176-191	BAC Water	1–2 mL per 2–5 mg	4–6 weeks	Same as AOD. BAC water standard.
IGF-1	0.6% Acetic Acid then BAC Water	0.5 mL AA first, then dilute to 1–2 mL	2–3 weeks	MUST use acetic acid. BAC water alone will cause aggregation and loss of potency. Critical.
IGF-1 LR3	0.6% Acetic Acid then BAC Water	0.5 mL AA first, then dilute to 1–2 mL	2–3 weeks	Same critical acetic acid requirement as IGF-1. Long R3 variant.
MGF (Mechano Growth Factor)	0.6% Acetic Acid then BAC Water	0.5 mL AA first, then dilute	2–3 weeks	Another IGF family member requiring acetic acid. Do not skip this step.
PT-141	BAC Water	1–2 mL per 10 mg	3–4 weeks	Can also be formulated as nasal spray: 10 mg in 10 mL = 1 mg/mL. Keep dark.
Melanotan I	BAC Water	1–2 mL per 10 mg	3–4 weeks	Light-sensitive. Store in amber vial or wrapped in foil. Keep cold.
Melanotan II	BAC Water	1–2 mL per 10 mg	3–4 weeks	Same as MT-I. Very light and heat sensitive. Never leave at room temp.
Epitalon	BAC Water	1 mL per 10 mg	4–6 weeks	Stable in BAC water. Khavinson protocol: 10-day course.
Na Epitalon	BAC Water	1 mL per 10 mg	4–6 weeks	Identical reconstitution to Epitalon.
MOTS-C	BAC Water	1–2 mL per 5–10 mg	4–6 weeks	Dissolves readily. Store reconstituted vial refrigerated.
SS-31 (Elamipretide)	BAC Water	1–2 mL per 10–20 mg	2–4 weeks	Light-sensitive. Keep in amber vial or wrapped in foil. Oxidation-sensitive.
FOXO4-DRI	BAC Water or 0.6% Acetic Acid	0.5–1 mL per 5 mg	3–4 weeks	Hydrophobic peptide. If cloudy in BAC water, use acetic acid first.
NAD+	Sterile Water (IV use — immediate)	Per clinical protocol (250–500 mg in 100–500 mL)	Use immediately for IV	IV: reconstitute in saline or sterile water immediately before infusion. Oral NMN/NR: stable powder — no reconstitution needed.
Thymosin Alpha-1	BAC Water	1 mL per 1.6 mg vial	3–4 weeks	Standard vial size is 1.6 mg. Dissolves readily.

Thymulin	BAC Water	1–2 mL per 0.1–0.5 mg	3–4 weeks	Low-dose peptide. Careful dilution math required for accurate dosing.
VIP	BAC Water or Sterile Saline	1 mL per 50 mcg	2–3 weeks	Very low dose (mcg range). Use precise dilution. Short half-life — split dosing common.
LL-37	BAC Water	1–2 mL per 1–5 mg	3–4 weeks	Can self-aggregate at high concentrations. Do not over-concentrate.
GHK-Cu / KPV Blend	BAC Water	1–2 mL per 5 mg blend	2–3 weeks	Combined vial. Use immediately or within 2–3 weeks refrigerated.
KPV (Oral)	Sterile Water (single use)	5 mL per 50 mg	24 hours (oral use)	For oral gut protocols: dissolve in plain sterile water. Drink immediately. Do not store.
PE-22-28	BAC Water	1 mL per 0.5–1 mg	3–4 weeks	Nasal spray use. 0.5–1 mg/mL concentration typical.
DSIP	BAC Water	1 mL per 0.5–1 mg	3–4 weeks	Sleep peptide. Very low dose (mcg range). Dilute accurately.
Oxytocin	BAC Water	1 mL per 0.5–1 mg (10 IU/mL typical)	2–3 weeks	Intranasal spray: 10 IU per 100 mcL spray. Degrades faster than most — use within 2–3 weeks.
Cerebrolysin	Ready to use (pre-formulated)	No reconstitution needed	Per product labeling	Supplied as aqueous solution. IV diluted in normal saline. Follow clinical protocol.
PNC-27	BAC Water or 0.6% Acetic Acid	1 mL per 1–5 mg	2–3 weeks	Experimental. Hydrophobic. Try BAC water first; acetic acid if cloudy.
Kisspeptin	BAC Water	1 mL per 0.1–0.5 mg	2–3 weeks	Very potent — micro-dosing typical. Use precise dilution protocol.
Abaloparatide	BAC Water	Supplied ready-to-use (FDA-approved pen)	Per device labeling	FDA-approved product. Supplied as a multi-dose injection pen. No reconstitution required.
Teriparatide	BAC Water	Supplied ready-to-use (FDA-approved pen)	Per device labeling	FDA-approved product. Pre-filled injection pen. No reconstitution required.

Red solvent text = requires acetic acid as first step. Green solvent text = BAC water is sufficient.

7. Step-by-Step Reconstitution Protocol

Follow this protocol every time you reconstitute a peptide vial. Sterile technique is non-negotiable.

1 — Gather supplies

Clean flat surface (alcohol-wiped). Peptide vial. BAC water vial (or acetic acid + BAC water). Two appropriately sized syringes (1 mL insulin syringe works well). Alcohol wipes. Gloves optional but recommended.

2 — Allow vial to warm

Remove the peptide vial from the freezer or refrigerator. Let it sit at room temperature for 15–20 minutes. This prevents condensation from entering the vial and prevents cold thermal shock to the powder when the solvent hits it.

3 — Wipe stoppers

Using a fresh alcohol wipe, clean the rubber stopper of both the peptide vial and the BAC water vial. Allow to air-dry for 30 seconds (do not blow on them).

4 — Draw your solvent

Insert a new syringe into the BAC water vial. Draw up the desired amount of BAC water (typically 1–2 mL for a 5 mg peptide vial). For acetic acid protocol: draw 0.5 mL of 0.6% acetic acid first.

5 — Inject slowly down the glass wall

Insert the syringe needle into the peptide vial at a slight angle so the tip touches the inside glass wall. Push the plunger SLOWLY. Let the solvent run down the glass wall and gently pool over the powder. Do NOT spray directly onto the powder. Do NOT inject rapidly.

6 — Do not shake

After adding solvent, set the vial down. Wait 2–5 minutes for the powder to absorb the liquid. Then gently roll the vial between your palms for 1–2 minutes. Never shake, vortex, or tap aggressively. The peptide should dissolve fully within 5–10 minutes.

7 — Check for clarity

A properly reconstituted peptide solution is typically clear and colorless. Some exceptions: GHK-Cu is blue-green (normal). Slight cloudiness may indicate aggregation — try warming slightly in your palms and rolling gently. Persistent cloudiness or particulates = do not use.

8 — Label and refrigerate

Write the peptide name, concentration (mg/mL), date of reconstitution, and your initials on the vial using a waterproof marker or Niimbot label. Refrigerate immediately. Do not leave at room temperature.

9 — Draw doses with new needle each time

Always use a new, sterile needle and syringe for every draw from the vial. Never reuse needles across draws. Wipe the stopper with a fresh alcohol wipe before every puncture.

8. Difficult, Gel-Like & Insoluble Peptides

Some peptides are notoriously difficult to dissolve. They may clump, form a gel, appear cloudy, or remain as visible powder even after extended mixing. This does not always mean the peptide is bad — it may simply require a different approach.

IGF-1 and IGF-1 LR3

Problem: Aggregates immediately in BAC water alone. Forms visible white strands or precipitate.

Solution: MUST use 0.6% acetic acid as the first solvent. Add 0.3–0.5 mL of 0.6% acetic acid, roll gently until dissolved (may take 10–20 min), then carefully add BAC water to desired final volume. Final solution should be clear. If still cloudy after acetic acid, warm to 37°C in a water bath for 5 minutes and roll again.

MGF (Mechano Growth Factor)

Problem: Same aggregation issue as IGF-1. Very hydrophobic C-terminal sequence.

Solution: Identical acetic acid protocol to IGF-1. No shortcuts. Always use 0.6% acetic acid first.

GHRP-6 and Hexarelin

Problem: Can appear slightly cloudy or form a white ring at the meniscus in BAC water.

Solution: Try BAC water first. If cloudy after 15 minutes of gentle rolling, switch to 0.6% acetic acid (0.5 mL), dissolve, then dilute with BAC water.

PNC-27

Problem: Hydrophobic experimental peptide. May not dissolve in BAC water at higher concentrations.

Solution: Start with 0.5 mL of 0.6% acetic acid. Roll for 20 minutes. Then dilute with BAC water. Keep concentration low (0.5–1 mg/mL maximum).

FOXO4-DRI

Problem: D-amino acid retro-inverso peptide. Unusual stereochemistry can cause solubility issues.

Solution: Try BAC water at low concentration (1 mg/mL). If cloudy, add a small amount of 0.6% acetic acid. Some researchers add 5–10% DMSO but this is not for injection use — topical only.

GHK-Cu (high concentration)

Problem: Blue-green color at low concentration is normal. At high concentration (>2 mg/mL) may become turbid due to copper complexation.

Solution: Keep concentration at 1–2 mg/mL. Do not over-concentrate. The blue-green color is the copper complex forming — this is expected and correct.

Tesamorelin (5 mg vials with variations)

Problem: Some batches dissolve instantly; others form a gel or slow-dissolving mass.

Solution: Add BAC water slowly down the glass wall. Do not rush. If gel-like after 10 minutes, warm the sealed vial in your palm at body temperature for 5 minutes and roll again. Tesamorelin is sensitive to pH — do not use acetic acid unless absolutely necessary.

SS-31 (Elamipretide)

Problem: Oxidation-sensitive. May develop a yellow tint if oxidized.

Solution: Use fresh BAC water. Work quickly to minimize air exposure. If the solution is yellow rather than colorless, the peptide may be degraded — yellow color is a warning sign, discard and test a new vial.

Kisspeptin

Problem: Very low dose (mcg range) makes concentration errors likely.

Solution: Use a 2-step dilution: first dissolve in 1 mL BAC water, then take 0.1 mL of that solution and add to 0.9 mL BAC water for a 10x dilution. This makes accurate dosing in the mcg range achievable with standard insulin syringes.

9. Freeze-Thaw Rules & Multi-Dose Vials

Freezing reconstituted peptide is possible and extends shelf life significantly. However, every freeze-thaw cycle stresses the peptide structure. The key is to freeze in single-use aliquots so you never thaw more than you need.

Maximum freeze-thaw cycles: Most peptides tolerate 3–5 freeze-thaw cycles without significant degradation. Beyond 5 cycles, expect meaningful potency loss. Beyond 10 cycles, the peptide should be considered compromised.

Aliquot before freezing: If you plan to freeze a reconstituted peptide, divide it into single-use aliquots immediately after reconstitution using insulin syringes or small vials. Each thawed aliquot is used completely and discarded — never re-frozen.

Thawing protocol: Remove from freezer. Allow to thaw at room temperature for 10–15 minutes. Do not microwave, heat, or run under warm water. Do not shake during thawing. Roll gently once thawed. Use within 4–6 hours once thawed.

Multi-dose vials with BAC water: BAC water vials with 0.9% benzyl alcohol can be used for multi-dose draws over 4–6 weeks without freezing, as long as proper needle technique is used each time. Wipe the stopper with a fresh alcohol wipe before every draw. Never reintroduce a used needle into the vial.

Never freeze glass ampoules: If your peptide came in a glass ampoule rather than a rubber-topped vial, do not freeze after opening — you cannot reseal it. Use the full contents immediately.

Date tracking: Label the date of reconstitution on every vial AND on each aliquot. If you cannot remember when you reconstituted it, throw it out.

10. Temperature Quick-Reference Card

FORM	TEMPERATURE	SHELF LIFE	KEY RULE
Lyophilized powder	-20°C (freezer)	1–2 years	Let warm to room temp before opening
Lyophilized powder	2–8°C (refrigerator)	3–6 months	Keep sealed. Use desiccant.
Lyophilized powder	Room temp (18–25°C)	Days to 2–4 weeks	Transit only. Not for storage.
Reconstituted (BAC water)	2–8°C (refrigerator)	4–6 weeks	Standard protocol. Wipe stopper every draw.
Reconstituted (BAC water)	-20°C (freezer)	3–6 months	Aliquot first. Max 5 freeze-thaw cycles.
Reconstituted (sterile water)	2–8°C (refrigerator)	5–7 days	Single-dose use only.
Reconstituted (any solvent)	Room temp (18–25°C)	24–48 hours max	Use immediately. Never store at room temp.

11. Warning Signs — When to Discard a Vial

■ When in doubt, throw it out.

Cloudiness or turbidity (that wasn't there before)	Precipitation, aggregation, or bacterial contamination. Do not use.
Visible particles or floating material	Bacterial growth, particulate contamination, or peptide aggregation. Do not inject.
Yellow or brown discoloration (unexpected)	Oxidation of sensitive peptides (SS-31, GHK-Cu at high dose). Degradation product. Discard.
Unusual smell when the stopper is punctured	Bacterial contamination or solvent breakdown. Discard.
Past shelf life date	Reconstituted vials past 6 weeks (refrigerated) should be considered suspect regardless of appearance.
Stopper looks discolored or corroded	Physical integrity of the vial is compromised. Do not use.

Lost the label	If you cannot identify the peptide, the concentration, or the date of reconstitution, discard it.
Any doubt whatsoever	Peptide vials cost money. Your health costs more. When uncertain, discard and reconstitute fresh.

12. Equipment & Supply Checklist

Everything you need for safe, accurate peptide reconstitution and storage.

Reconstitution Supplies

- Bacteriostatic Water (BAC water) for injection — 30 mL multi-dose vial
- Sterile Water for Injection — single-use ampoules (for immediate single-dose use)
- 0.6% Acetic Acid solution (for IGF-1, LR3, MGF, and difficult peptides)
- 1 mL insulin syringes (U-100, 28–31 gauge needles) — one per draw minimum
- 3 mL syringes for drawing BAC water
- Alcohol wipes (70% isopropyl alcohol)
- Sterile gloves (optional but recommended)
- 0.22 micron syringe filters (for filtering acetic acid before use)

Storage Equipment

- Amber or dark glass vials (if transferring to light-protected container)
- Small airtight container or pill organizer for vial storage
- Silica gel desiccant packets
- Dedicated section of refrigerator (not the door — temperature fluctuates at door)
- Small freezer-safe container for long-term storage
- Permanent marker or Niimbot M2 label printer + labels
- Parafilm or PTFE tape for sealing unreconstituted vials

Optional but Recommended

- PeptiCool or equivalent peptide travel cooler (for transport)
- Sharps disposal container (never throw needles in household trash)
- Logbook or tracking spreadsheet: peptide, date reconstituted, concentration, remaining volume
- Small digital scale for verifying vial weights if suspected underfill
- Thermometer for verifying refrigerator and freezer temperatures

Safety Essentials

- Alcohol wipes — wipe every stopper before every needle puncture
- New sterile needle and syringe for every draw
- Sharps container within arm's reach of work area
- Never use needles on more than one vial without changing them
- Work on a clean, flat surface wiped with 70% isopropyl alcohol

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